



Pyrolysis of 2-methylfuran/jet fuel surrogate blends: An experimental and kinetic modeling study

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ABSTRACT

Owing to significant advantages over traditional biofuels, 2-methylfuran (MF) has arisen extensive interest as next generation biofuels. Atmospheric pressure pyrolysis of a 1:1 blend of MF with a three-component (3C) surrogate jet fuel was conducted in a flow tube using synchrotronphotoionization and molecular beam mass spectrometry techniques within a temperature range of 898–1223 K. Therein, the 3C surrogate fuel consists of 66.2% n-dodecane (NC₁₂H₂₆), 18.0% 1,3,5-trimethylcyclohexane (T135MCH) and 15.8% n-propylbenzene (NPB, A1C₃H₇). The detailed kinetic model including 579 species and 3474 reactions was developed via validating against both measured and literature data with reasonable agreement. MF exhibits no obvious influence on the 3C components pyrolysis. However, 3C mixtures play an important roles in MF decomposition: lower initial temperature and different decomposition pathways. In the meantime, n-propylbenzene (NPB) pyrolysis curve in 3C and 3C+MF significantly transfers to low temperature zone after 1000 K in contrast to pure NPB, while smaller change was found in the decomposition curve of n-dodecane. Compared with 3C pyrolysis, 3C+MF blends have less formation of aromatic species and light hydrocarbons except for vinylacetylene (C₄H₄) and acetylene (C₂H₂). Allyl radical (C₃H₃) has significant roles in benzene ring formation. The combination of allyl radical with propargyl radical producing benzene and H radical (aC₃H₅ + C₃H₃ = A1 + 2H) has the highest contribution to the benzene formation in both 3C and 3C+MF. Moreover, most of naphthalene (A2) formation is also related to C₃H₃ radical via reaction: A1CH₂ + C₃H₃ = A2 + H, rather than via HACA (hydrogen abstraction carbon addition) reaction. Interaction of fuel blends was mainly presented through the common and important sensitive reactions which are much to do with CH₃ and H radicals. The overview of product distribution was investigated by providing comparisons of carbon conversion rates during the pyrolysis of 3C and 3C+MF. These results will be helpful to better understand the pyrolysis properties of MF/jet fuel surrogate blends and could contribute to developing new generation biofuels for aviation.

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1. Introduction

Energy is fundamental power to promote the rapid and sustainable development of the global social economy. Nevertheless, excessive consumption of petroleum, caused by the rapid development of industrialization and modernization, results in higher emissions of pollutants as well as greenhouse gas (GHG). Developing sustainable alternative energy, such as solar, nuclear, and bio-fuel will be a useful way to reduce environmental pollutions, and

relieve the case of low fossil fuel storage. In recent years, oxygenated fuels have been widely recognized as the most potential biofuels.

Mixtures of hydrocarbons and oxygenated fuel have been proved to achieve a good inhibition effect on soot formation [1–10]. Ethanol, one of the most welcome oxygenated fuel, has been used as a common alternative fuel in the transport sector due to its mature production mode [11–13]. However, many disadvantages limited its large scale application, for example, high latent vaporization heat, low energy density, and water solubility. Besides, bioethanol extraction from crops will threaten food security due to the large population and the limited cultivated land area. Furan-based fuels, generated from non-edible feedstocks, es-

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Table 1
Fuel properties [1,2,7,9–11, 15,19–24].

Fuel	Gasoline	Diesel	Bioethanol	MF	RP-3
Chemical formula	C ₂ –C ₁₄	C ₁₂ –C ₂₅	C ₂ H ₆ O	C ₅ H ₆ O	C _{10.57} H _{21.99}
Density at 20 °C (kg/m ³)	745	826	791	910	783
Viscosity (mPa s)	~0.42	2.86	1.08	4.00	1.72
Water solubility at 20 °C (wt%)	Negligible	Negligible	Miscible	0.3	Negligible
Research octane number	96.8	20–30	109	100.7	–
Motor octane number	85.7	–	90	86	–
Cetane number	10–15	52.1	8.0	8.3	41.5
Initial boiling temperature (°C)	32.8	180–370	78	64	–
Latent heat of vaporization (kJ/kg) at 25 °C	373	270–301	920	358	291
Stoichiometric air requirement (l)	14.7	14.3	9.0	10.1	–
Lower heating value (MJ/kg)	42.9	42.5	26.9	30.4	42.8
Auto-ignition temperature (°C)	420	180–220	434	–	–
H/C ratio	1.795	1.91	3	1.2	2.0
Gravimetric oxygen content (%)	0	0	34.78	19.51	0

pecially 2-methylfuran (MF) and 2,5-dimethylfuran (DMF), exhibited additional benefits as next generation biofuels due to their significant advantages. MF has many advantages compared with bioethanol [1,14]. The insolubility of MF in water could make it as well as its blends with other fuels more stable. The latent heat of vaporization for MF (358 kJ/kg) is much smaller than that for bioethanol (920 kJ/kg), which could prevent the engine cold-start problem. Furthermore, higher energy density for MF (30.4 MJ/kg) against bioethanol (26.9 MJ/kg) could decrease fuel consumption and improve engine efficiency. Combustion characteristic test in spark ignition (SI) engines [1,9,15,16] indicates that MF has a higher research octane number (RON), higher energy density, and lower hydrocarbon emissions by comparison with gasoline and ethanol. Table 1 also presented some properties of gasoline, diesel, bioethanol, MF and jet fuel (RP-3). Moreover, previous literature [17,18] has made the breakthrough of furan-based production methods, which could accelerate the process of their industrial application.

There is already some literature available focusing on MF combustion properties and its blend with gasoline/diesel/biodiesel [1,7–10, 15,25,26]. Wang et al. [1] applied pure components (MF, DMF, ethanol) as well as gasoline into direct injection compression ignition (DIC) engine to compare their combustion characteristics and emissions under specific conditions. The knock suppression ability of MF is similar to DMF and superior to gasoline. MF could result in lower volumetric fuel consumption in contrast to ethanol. Pan et al. [27] conducted a comparison of combustion performance between MF and gasoline within the spark-ignition engine. During MF combustion procedure, it could generate higher cylinder pressure, combustion temperature, knock intensity and nitrogen oxide (NO_x) emissions than gasoline at a high compression ratio. Xiao et al. [7,8,10] investigated the combustion performance of MF/diesel blends as well as MF/biodiesel blends at different blending ratios within the DIC engine. It was found that MF addition in diesel and biodiesel can reduce the soot emissions at most experimental conditions, while it also can enlarge the NO_x emissions. The blends exhibit shorter combustion duration and longer ignition delay times in contrast to pure diesel and biodiesel. However, they indicated that MF was comparable to other fuels from the side of the overall regulated emissions, and aldehyde emission from MF is less than gasoline and bioethanol. Works from Aachen University [15,28] focused on the combustion behavior of pure fuel molecules (MF, butanone, ethanol) and RON95 gasoline in DIC engine. They found that MF has higher combustion efficiency, better knock resistance as well as excellent combustion stability. Moreover, MF could reduce significant particle emissions compared to conventional gasoline. When compared to ethanol, MF produces higher emissions of nitrogen oxides. Tripathi et al. [25] compared the oxidation performance of MF and its blend with n-heptane

from the experimental and kinetic side. MF/n-heptane blends have a shorter ignition delay time compared with pure MF. They also proposed that the main interaction between MF and n-heptane is related to smaller radicals, especially OH.

Owing to the complex components for real aviation fuel, surrogate fuel is applied to represent the aviation fuel, studying their combustion properties. Given average relative molecular mass, proportion of hydrocarbons, C/H ratio as well as relative error of ignition delay in RP-3 fuel, a three-component (3C) fuel: 66.2% n-dodecane (NC₁₂H₂₆), 18.0% 1,3,5-trimethylcyclohexane (T135MCH) and 15.8% n-propylbenzene (NPB, A1C₃H₇) was proposed as RP-3 surrogate fuel by Xu et al. [29]. Furthermore, Liu et al. [30] compared ignition delay times and laminar burning velocities between 3C surrogate fuel and Jet A-1 experimentally, and drew the conclusion that they have similar combustion properties. In addition, they recommended 3C is appropriate to be surrogate fuel of RP-3 and a Jet A-1, considering their composition, C/H ratio of 3C as well as its combustion behavior. Previous literature [30,31] has systematically studied the combustion behavior of 3C surrogate fuel experimentally and kinetically, involving its pyrolysis, oxidation, flame speed and ignition delay times. Many studies focused on the combustion properties of MF as well as its blends with vehicle fuels. So, it is important to investigate the combustion characteristics of MF/aviation fuel blends. However, comparison of combustion properties between aviation fuels and MF/aviation fuels is rare.

The present work aims to provide insight into pyrolysis properties of 3C/MF blends (3C+MF) experimentally and theoretically as well as comparing its pyrolysis behavior with 3C. 3C+MF pyrolysis experiment was conducted in a flow reactor under atmospheric pressure from 898 to 1223 K. Synchrotron radiation photoionization and molecular beam mass spectrometry (MBMS) techniques were used to achieve measurements of pyrolysis products. Then, an overview of product distribution was investigated by providing a comparison of carbon conversion ratio during the pyrolysis of 3C and 3C+MF. A comprehensive kinetic model was developed to give kinetic analysis of their pyrolysis properties by validating against species profiles. A comparison of pyrolysis characteristics among 3C+MF blends, 3C surrogate fuel as well as pure component fuel is also presented.

2. Experimental

Pyrolysis of 3C+MF fuel was investigated at National Synchrotron Radiation Laboratory (NSRL) in Hefei, China. Previous literature [32,33] has provided the details of the beamlines used within the present work. In short, the photon energy can be selected from 7.5–22.0 eV, and its energy resolution is 3000 (E/ΔE at 10 eV). The beamlines could reduce fragmentation from near-threshold photoionization as much as possible with easy tunability

Table 2
Composition of gas mixtures (%).

Fuel mixtures	MF	NC ₁₂ H ₂₆	T135MCH	NPB	Ar
3C+MF	0.500	0.331	0.090	0.079	999.000
3C	0.000	0.662	0.180	0.158	999.000

and high energy resolution. Pyrolysis setup is composed of three major parts: a) pyrolysis chamber, b) molecular-beam sampling system, c) photoionization chamber equipped with homemade reflection time-of-flight mass spectrometer (RTOF-MS). The flow reactor (diameter 7 mm, length 400 mm), made of α -Al₂O₃, locates in the center of the pyrolysis chamber. Its centerline temperatures, given in the supplemental material (SM), were measured by an S-type thermocouple. Uncertainty of temperature profiles is supposed to be ± 30 K. It should be noted that the longer length of flow reactor (400 mm) could provide longer high temperature zone and reaction time in contrast to previous works [34–36]. A quartz nozzle at the outlet of the flow reactor was used to sample the pyrolysis products, forming a molecular beam that can be ionized by synchrotron vacuum ultraviolet light to produce charged ions. Ion signals can be detected via a microchannel plate detector and recorded by a multiscaler (P7888, FAST Comtec, Germany).

Inlet fuel components are as follows: NC₁₂H₂₆ (>98%, Aladdin), T135MCH (>98%, TCI, Japan) and NPB (>99%, TCI, Japan), MF (>98%, Aladdin). The composition of mixtures that are used within of experiment of 3C and 3C+MF, were shown in Table 2. In the present work, the composition of 3C+MF is a 1:1 blend of MF with the 3C surrogate jet fuel. The mixture of 3C+MF is vaporized with a set temperature of 498 K, then carried by argon (99.999% purity). The total inlet gas flow is 1000 standard cubic centimeters per minute (sccm). Pyrolysis pressure is controlled at 760 Torr by an MKS throttle valve. Previous work [37] has reported means of intermediate identification and mole fraction evaluation in detail. The experimental uncertainty of major species is supposed to be $\pm 10\%$, that of species, with known photoionization cross sections (PICS), are $\pm 25\%$. As for species with estimated PICSs, their experimental uncertainty is supposed to be a factor of two. PICS can be gotten from the database [38]. Carbon balances are closed within 10% within experimental temperature points. Carbon balances, as well as experimental data, are available in SM.

3. Kinetic model

The pyrolysis of 3C and 3C+MF studied within the plug flow reactor was simulated using the PFR code of Chemkin-Pro [39]. The present kinetic mechanism, developed based on the previous model for 3C surrogate fuel [31], includes 579 species and 3474 reactions. Mechanism and thermodynamic data are available in SM. Sub-mechanisms for n-dodecane, T135MCH and NPB have been introduced in detail in the 3C surrogate fuel model [31], which has been validated by species profiles in the flow tube as well as Jet-stirred reactor (JSR) with good agreement. Briefly, the sub-model of n-dodecane has included unimolecular C–C and C–H fission reactions and H abstraction reactions with radicals, referring to previous kinetic models [34,40,41]. The sub-model of T135MCH considers demethylation reactions, H abstraction reactions and ring-opening reactions, rate constants of these reaction types come from the literature [42,43] on kinetic studies for cyclic hydrocarbons. Owing to the stability of benzene ring structure, The NPB sub-model mainly considers C–C and C–H bond breakage at the alkyl chain [30,44,45]. There is already much literature studying pure MF pyrolysis from the experimental and kinetic side [46–53]. In the present sub-model of MF, H abstraction reactions at methyl side referred to Somers et al. [53] (with H radical), Liu et al. [54] (with CH₃ radical), as well as Cheng

et al. [47] (with C₃H₃, aC₃H₅, nC₄H₅, etc.). As the most important decomposition pathways within pure MF pyrolysis, unimolecular reactions in the present model were taken from previous literature [49]. Moreover, the present model also includes H addition reactions of MF, referring to the mechanism reported by Cheng et al. [47]. Furthermore, the coupling reactions among MF and other fuel-derived radicals, with 12 reactions involved, were newly considered by the present work by referring to mechanism [47] developed for MF pyrolysis. The dehydrogenation reaction of MF with CH₃ radical on the methyl side is considered as analogy reaction to determine the rate expressions of the above reactions. Six n-dodecane radicals (PXC₁₂H₂₅, SXC₁₂H₂₅, S2XC₁₂H₂₅, S3XC₁₂H₂₅, S4XC₁₂H₂₅ and S5XC₁₂H₂₅), three NPB radicals (A1CHCH₂CH₃, A1CH₂CH₂CH₂ and A1CH₂CHCH₃), three T135MCH radicals (SXD35MCH, PXCH2D35MCH and TXT135MCH), were introduced into coupled reaction network. Previous works [30,31] have already considered the coupling reactions among the 3C component. Thermochemical data are mainly from the core model [31]. As for species involving MF decomposition, they are taken from Cheng et al. [47].

4. Results and discussion

By using RTOF-MS, 24 species profiles including CO, light hydrocarbons and aromatic species were determined under pyrolysis process of 3C+MF. Experimental results from 3C surrogate fuel pyrolysis [31] were also introduced in the present study, for the sake of assessing the quality of mechanism as well as studying the effect of MF addition on 3C surrogate fuel. Simulations for the pyrolysis of 3C and 3C+MF were performed using the present model. Furthermore, a comparison of fuel decay behavior among 3C, 3C+MF and single component fuel was made to study their different consumption characteristics. Therein, simple rate-of-production (ROP) of MF consumption was conducted for comparing its difference with pure MF on decay behavior. The carbon conversion ratio of 3C and 3C+MF was investigated by comparing carbon content in light hydrocarbons and set of aromatic.

4.1. Fuel decomposition

The fuel conversion ratio in Fig. 1 was made to compare the fuel decomposition characteristic at different temperature points. 3C+MF data were measured by present work, 3C data were taken from Jin et al. [31]. It should be noted that the inlet mole concentration of fuel is 1 % both in 3C and 3C+MF. Pure single component data for NC₁₂H₂₆, NPB and MF were obtained from Zeng et al. [34], Yuan et al. [35] and Cheng et al. [47], respectively. Because there is no reported literature data for T135MCH pyrolysis, the comparison for pure T135MCH could be done in further work. The inlet concentration of pure components (NC₁₂H₂₆, NPB and MF) is 2%, 1%, 2%. It is worth to be mentioned that all the above experimental data were detected by RTOF-MS in NSRL under atmospheric pressure. The material and diameter of the flow tube are same in the mentioned above experiments. The length of the flow reactor is 400 mm within 3C and 3C+MF experiments. As for pure single components, their lengths are 150 mm (NPB and MF) and 220 mm (NC₁₂H₂₆). Hence, the residence times of the diameter of 400 mm, 220 mm, 150 mm are 0.015 s, 0.008 s, 0.006 s under atmospheric pressure at 298 K.

As displayed in Fig. 1, there are very similar decay tendencies between 3C and 3C+MF for NC₁₂H₂₆, NPB, T135MCH, even with different inlet fuel concentrations. The initial decomposition temperature of pure NPB is closed to that of 3C and 3C+MF. However, NPB pyrolysis curve in 3C and 3C+MF transfers to low temperature zone after 1000 K. The main cause of the above “NPB

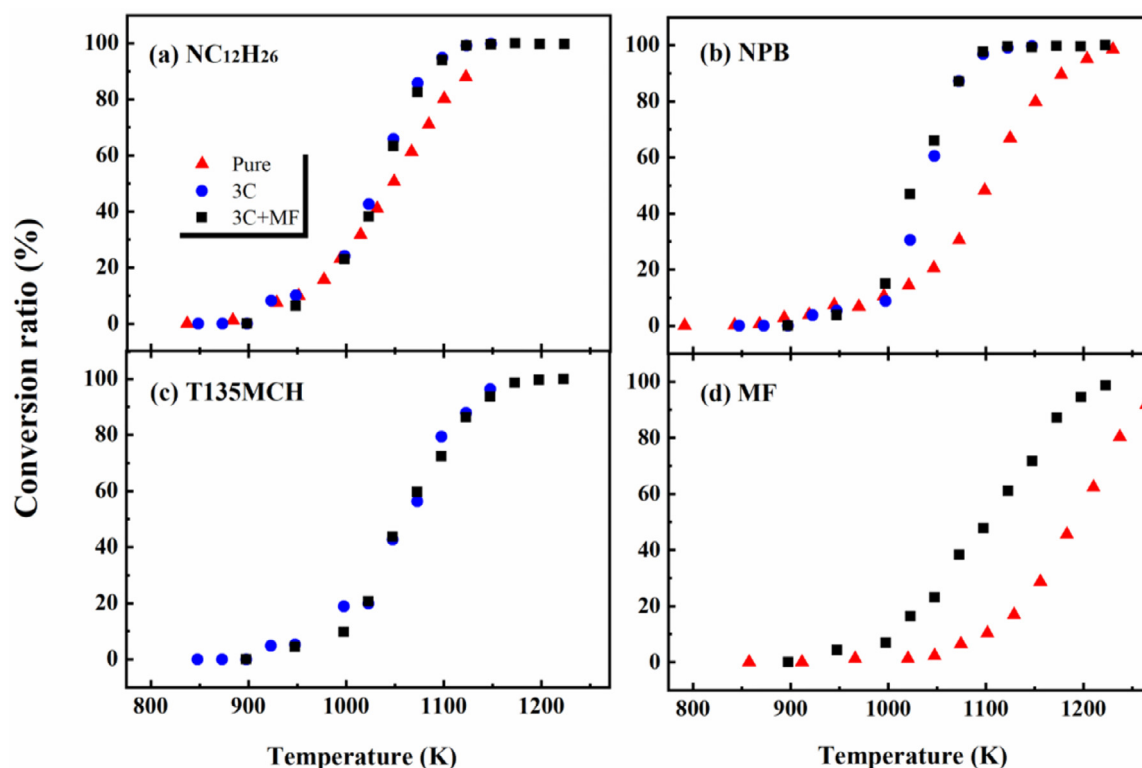


Fig. 1. Comparison of fuel conversion ratio. The 3C+MF data were detected by this work. 3C pyrolysis data comes from Jin et al. [31]. Pure single component data for $\text{NC}_{12}\text{H}_{26}$, NPB and MF were obtained from Zeng et al. [34], Yuan et al. [35] and Cheng et al. [47], respectively.

gap” is that pure NPB experiment cannot provide longer residence time, especially at high temperature compared to 3C and 3C+MF experiment. In addition, NPB was mainly consumed via dissociation of the benzylic C–C bond ($\text{A1C}_3\text{H}_7 = \text{A1CH}_2 + \text{C}_2\text{H}_5$) based on ROP performed in the present and previous work [35]. A1CH_2 , the direct product of dissociation reaction, could easily be consumed via binding reactions with H and CH_3 radicals ($\text{A1CH}_2 + \text{CH}_3 = \text{A1C}_2\text{H}_5$ and $\text{A1CH}_2 + \text{H} = \text{A1CH}_3$). Therefore, abundant H and CH_3 radicals, from other $\text{NC}_{12}\text{H}_{26}$ decay, can promote the consumption of NPB and A1CH_2 in 3C and 3C+MF, which could also contribute to the “NPB gap”. $\text{NC}_{12}\text{H}_{26}$ exhibits a limited “gap” compared to a pronounced one in NPB. One reason is that length of the flow tube in pure $\text{NC}_{12}\text{H}_{26}$ pyrolysis is longer than pure NPB pyrolysis. Another one could be attributed to as follow: the most important decay pathway of $\text{NC}_{12}\text{H}_{26}$ is the H abstraction reaction. However, other fuels and their derived molecules will compete with $\text{NC}_{12}\text{H}_{26}$, especially for H radical. The limited “gap” in $\text{NC}_{12}\text{H}_{26}$ could be the result of different residence times at high temperatures and competition for H radical.

MF pyrolysis curve in 3C+MF blends leads toward low temperature zone than a pure MF curve, even for initial temperature. Simple ROP analysis for MF was performed to study the “MF gap” at 1123 K (61.0% conversion) and 1173 K (87.1% conversion). Figure 2 includes more than 90% MF decay pathway, and other unimportant consumption pathways have been discussed in detail in previous literature [46,47,49]. According to ROP for MF pyrolysis performed by Cheng et al. [47], H migration reaction, producing cyclic carbenes (pathway 1 and 3), are the most important decay pathways during pure MF pyrolysis. Moreover, H abstraction reactions play limited roles in pure MF consumption. However, Fig. 2 indicates that the H abstraction reactions at the methyl side are dominant pathways for MF decomposition in present work (pathway2) as a result of longer residence time at high temperature and the abundant radical pools (especially for the higher concentration of H rad-

ical) in 3C+MF experiment compared to pure MF decay. Moreover, MF has no obvious effect on the consumption behavior of 3C components. On the contrary, the latter could make the decomposition of MF shift to the low temperature zone.

As presented in Fig. 3, the present mechanism could reproduce experimental results with good agreement for both 3C and 3C+MF. Except for MF, there is no big difference for important fuel decay pathways among in 3C, 3C+MF and pure component fuel according to the ROP analysis performed by the present model. Consumption pathways for MF have been discussed in the above paragraph. Previous literature [31,34,35,41,55–57] has discussed the decomposition pathways for $\text{NC}_{12}\text{H}_{26}$, T135MCH and NPB in detail. Their main decomposition pathways are presented briefly in 3C+MF here. Furthermore, their detailed decomposition pathways are displayed in SM. Most of the $\text{NC}_{12}\text{H}_{26}$ decay underwent H abstraction reactions via radicals, especially H radical. Unimolecular C–C fission reactions have some contributions for $\text{NC}_{12}\text{H}_{26}$ consumption too. NPB mainly decayed via reaction $\text{A1C}_3\text{H}_7 = \text{A1CH}_2 + \text{C}_2\text{H}_5$. H abstraction reactions at the chain side also play important roles in NPB consumption. Ipso-addition reaction by the H radical slightly contributes to the consumption of NPB. As for T135MCH, its major decomposition channel is the dehydrogenation reaction with H atom. Demethylation reaction also contributes high a percentage of fuel decay. C–C bond fracture on ring occupied less than 10% of T135MCH consumption.

4.2. Sensitivity analysis

Sensitivity analysis was conducted at 1173 K based on the present model to study the coupling effect of 3C+MF. Mixture decay ($\text{NC}_{12}\text{H}_{26}$: 99.6% conversion; NPB: 98.6% conversion; T135MCH: 99.9% conversion; MF: 87.1% conversion) is close to finished within this temperature point, so it is easy to find sensitive reactions of fuel consumption correctly. Sensitivity analysis for individual

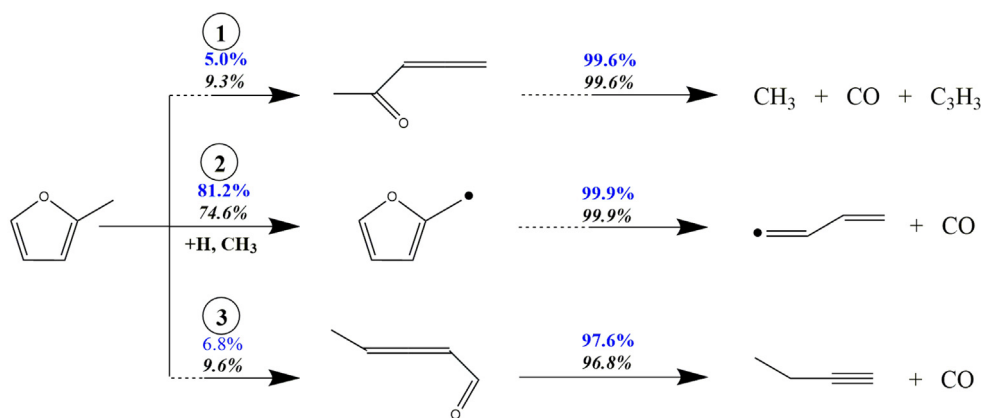


Fig. 2. Simple ROP analysis for MF decomposition at 1123 K (61.0% conversion) and 1173 K (87.1% conversion, italic).

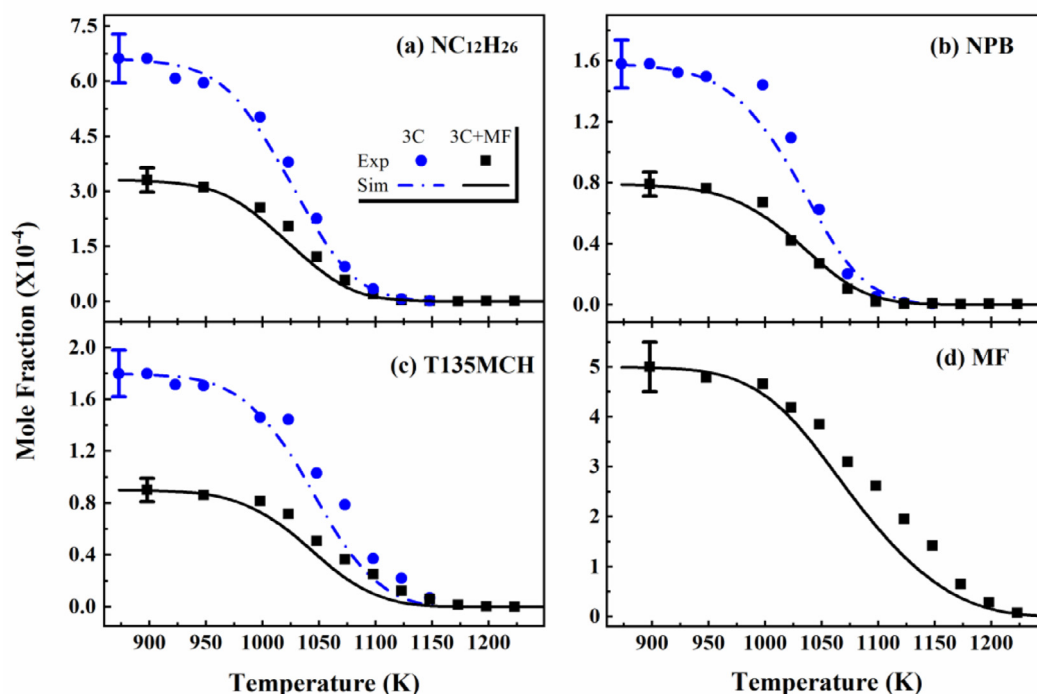


Fig. 3. Experimental (symbols) and simulated (lines) mole fraction profiles of $\text{NC}_{12}\text{H}_{26}$ (a), T135MCH (b), NPB (c) and MF (d). 3C pyrolysis data comes from Jin et al. [31].

fuel can be found in Section 4 of SM. Reactions involving fuel molecules, play important roles in individual fuel decomposition. Coupling reactions between fuels and fuel-derived radicals are insensitive to the consumption of fuel blends. Common and important sensitivity reactions in Fig. 4 are almost related to small molecules, especially related to CH_3 and H radicals which are important for fuel consumption. Therein, only R4 shows the inhibiting effect for all inlet fuel molecules, resulting from its higher consumption of H radical. It can be found that the sensitivity of small molecule reactions to NPB decomposition exhibits different tendencies compared with other fuel molecules. For example, R1 performs promoting effect for $\text{NC}_{12}\text{H}_{26}$, T135MCH and MF, but inhibits NPB decomposition. According to ROP, R1 mainly occurs via reverse reaction, which would produce H, but consume CH_3 . The inhibiting effect of R1 is attributed to that R1 can compete with A1CH_2 for CH_3 radical, while the main A1CH_2 decomposition pathway is to combine with CH_3 radical to produce ethylbenzene ($\text{A1C}_2\text{H}_5$). The R1 promoting effect can be explained as follow: H radical could greatly promote fuel consumption through H abstraction reactions. Therefore, NPB decay is more sensitive to CH_3 than H,

while $\text{NC}_{12}\text{H}_{26}$, T135MCH as well as MF prefer H radical, which also has been confirmed by previous work [31]. The above analysis could also explain the same phenomena for R3 and R5. As for R2, its promoting effect for $\text{NC}_{12}\text{H}_{26}$, T135MCH and MF is owing to its product C_2H_5 which is the major source of H atom based on ROP analysis. R6 has positive factors for $\text{NC}_{12}\text{H}_{26}$, T135MCH and MF due to H radical consumption, while its limited negative factor is caused by C_3H_6 which can produce CH_3 radical via R3.

4.3. Major species

As presented in Fig. 5, the present model could reproduce measured results of major species with reasonable agreement. Based on ROP for MF in Fig. 2, all the major decomposition pathways lead to CO eventually. Therefore, CO exhibits a high concentration in the present work, which is consistent with the pure MF decomposition experiment [47]. However, other stable oxygenated products such as $\text{C}_4\text{H}_4\text{O}$ (furan + 1-oxo-1,3-butadiene) appear with low concentration, their simulation results are displayed in SM. Although there is an oxygen atom in the MF molecule, about 80%

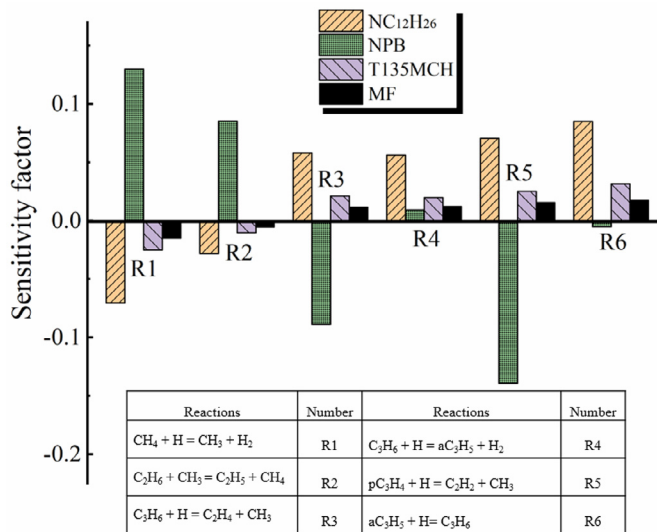


Fig. 4. Common and important sensitivity reactions for 3C+MF pyrolysis at $T = 1173$ K (NC₁₂H₂₆: 99.6% conversion; NPB: 98.6% conversion; T135MCH: 99.9% conversion; MF: 87.1% conversion).

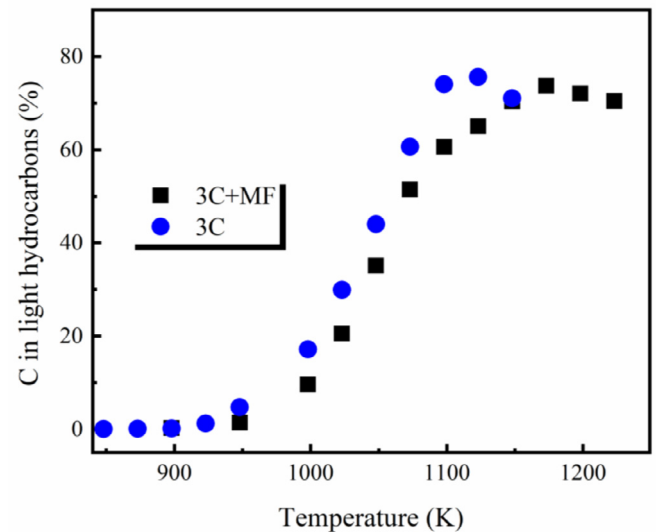


Fig. 6. Comparison of carbon conversion rate between 3C [31] and 3C+MF in view of light hydrocarbons.

of the O atom is released in the form of CO easily after 1150 K, which indicates there are small amounts of aldehydes and ketones during the pyrolysis process of MF, based on the present experimental and literature results [47]. Compared with 3C pyrolysis, 3C+MF has a lower concentration of light hydrocarbons except for vinylacetylene and acetylene. Therein, experimental data of acetylene in 3C pyrolysis is multiplied by 2 for a better view. According to the ROP analysis in Fig. 2, the pathway 2 is the most impor-

tant way for MF decay. nC₄H₅, main product of pathway 2, undergoes two major decomposition pathways: (A) nC₄H₅ = H + C₄H₄, (B) nC₄H₅ = C₂H₂ + C₂H₃. In 3C+MF pyrolysis, reaction A accounts for most of the C₄H₄ generation. Reaction B can contribute a lot of acetylene formation directly. Furthermore, C₂H₃, the product of reaction B, is also the major source of acetylene through reaction: C₂H₃ = C₂H₂ + H. Therefore, MF could result in higher concentrations of C₂H₂ and C₄H₄. Acetylene and ethylene are the

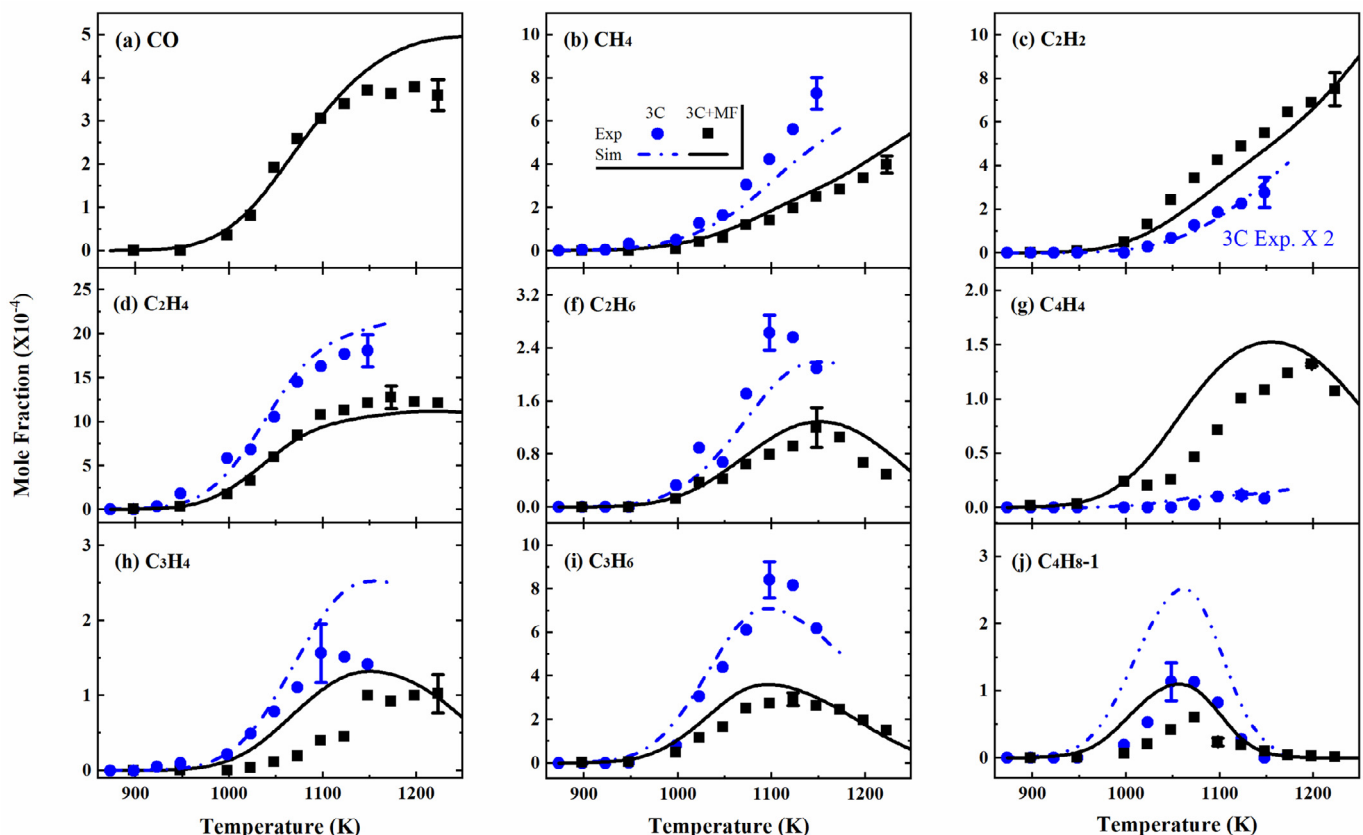


Fig. 5. Experimental (symbols) and simulated (lines) mole fraction profiles of major species. 3C pyrolysis data comes from Jin et al. [31].

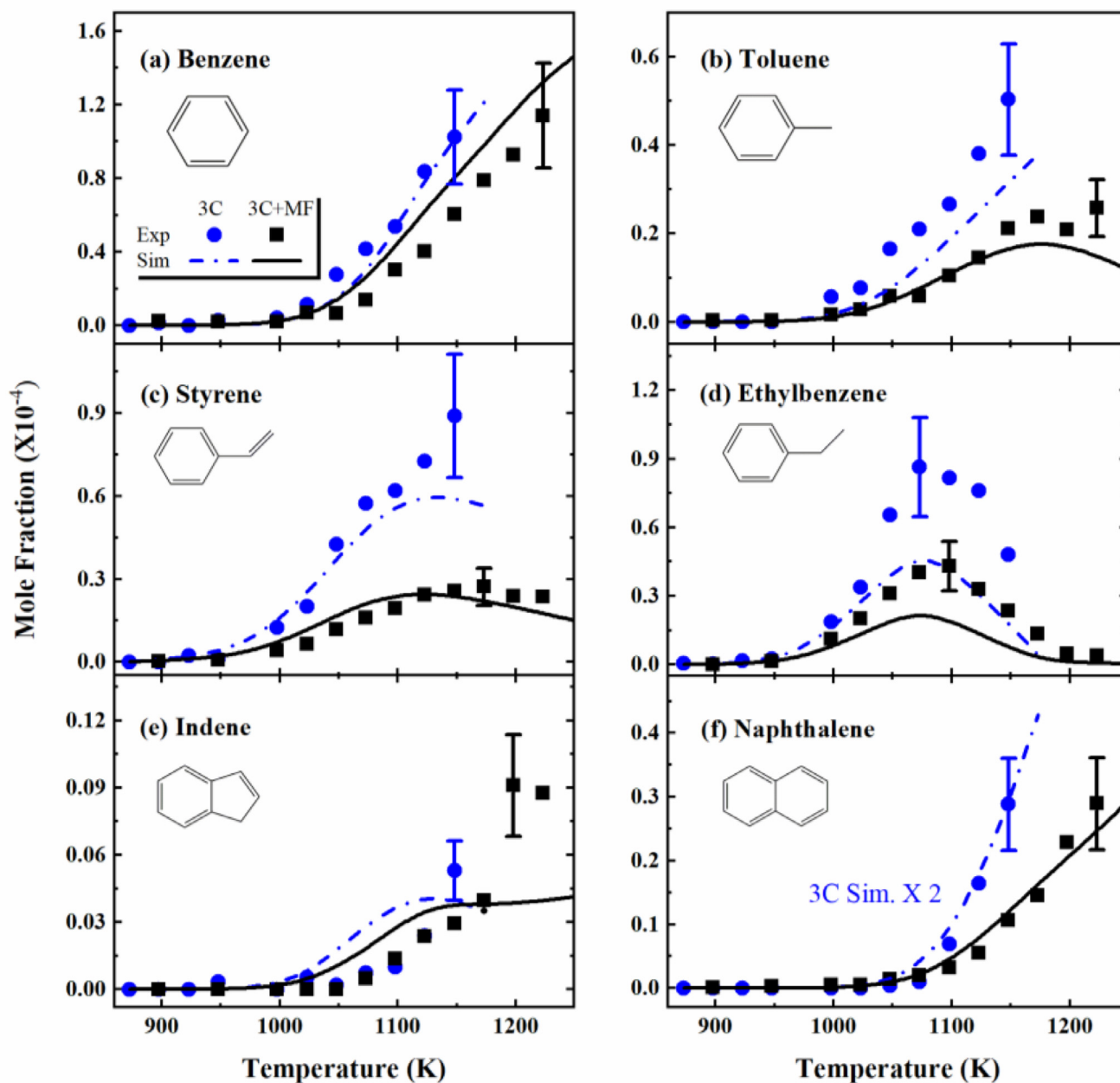


Fig. 7. Experimental (symbols) and simulated (lines) mole fraction profiles of major species. 3C pyrolysis data comes from Jin et al. [31].

most important products in 3C+MF. During the pyrolysis of 3C and 3C+MF, most ethylene results from the β -C-C scission reactions of some radicals (nC_3H_7 , pC_4H_9) which stems from sequential decomposition reactions of $C_{12}H_{25}$ radicals. Other alkenes (propene, C_3H_6 ; butene, C_4H_8-1) are almost related to the decay of $C_{12}H_{25}$ radicals too. As for C_3H_4 (propyne and allene), it mainly comes from allyl radical (aC_3H_5) which is the result of dehydrogenation of propene as well as decomposition alkenes with a higher carbon. Therefore, the lower concentrations of light hydrocarbons (except for vinylacetylene and acetylene) observed in 3C+MF pyrolysis could come from lower inlet concentrations of the 3C component, especially $NC_{12}H_{26}$.

Carbon inlet mole concentrations are different between 3C (carbon inlet concentration: 10,990 ppm) and 3C+MF (carbon inlet concentration: 7990 ppm), even with the same total inlet mole concentrations of two mixtures. Moreover, 3C+MF also has

500 ppm oxygen input, with a C/O ratio of 15.98. Owing to no oxygen in 3C, hence, the carbon conversion rate was chosen to compare the product distributions (light hydrocarbons and aromatic species) between 3C and 3C+MF. Figure 6 shows the conversion ratio of carbon in imported fuel to carbon in light hydrocarbons at observed experimental temperatures. Carbon in light hydrocarbons is the sum of carbon concentration in all observed light hydrocarbons. In beginning, the carbon conversion rate always increases with temperature, the conversion rate in 3C is slightly higher than that in 3C+MF at the same temperature point. Peak values of conversion rate are about 75% in 3C and 3C+MF. The corresponding peak temperature of 3C is 1123 K, which shifts to 1173 K for 3C+MF. Moreover, the carbon conversion rate begins to decrease at the last few points both in the pyrolysis of 3C and 3C+MF. According to the species profiles of light hydrocarbons, it could be attributed to the decrease of C_3H_4 and C_3H_6 . As mentioned

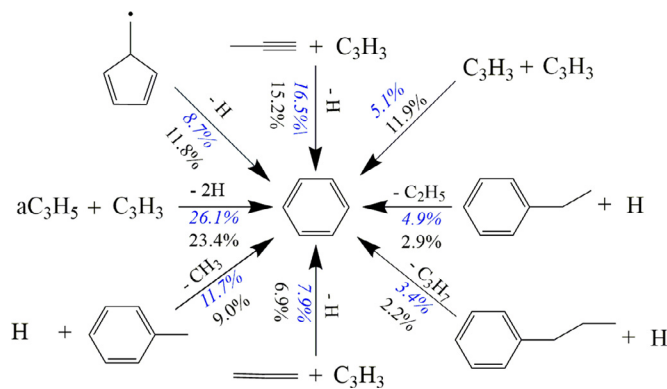


Fig. 8. Benzene formation pathway at 1173 K (3C: italic).

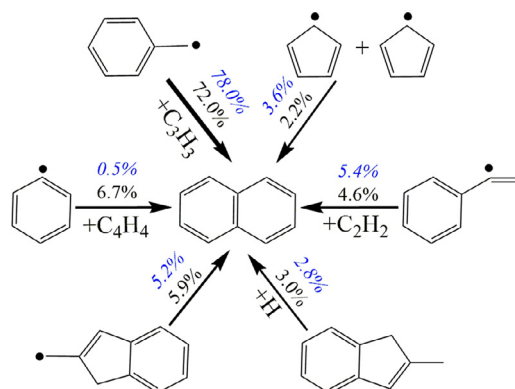


Fig. 9. Naphthalene formation pathway at 1173 K (3C: italic).

above, about half of C_3H_6 could convert into C_3H_4 . As presented in Section 4.4, C_3H_4 could easily produce C_3H_3 radical which is the important source of benzene. Hence, the carbon reduction in light hydrocarbons finally comes into the set of aromatics.

4.4. Aromatic species

Figure 7 presents species profiles of aromatic species measured within this work. Simulated results achieve general agreement with measured data. The simulation result of naphthalene (A2) in 3C is multiplied by 2 for a better view. The observed concentration of the set of aromatics in the 3C experiment is generally higher than that in the 3C+MF experiment. Toluene (A1CH_3) formation is related to A1CH_2 which is the important product of NPB decomposition both in 3C and 3C+MF experiments. Reaction $\text{A1CH}_2 + \text{H} = \text{A1CH}_3$ contributes most of toluene formation pathway (86.6% contribution in 3C+MF, 76.3% contribution in 3C). Ethylbenzene ($\text{A1C}_2\text{H}_5$) also is an important product directly related to A1CH_2 radical. Almost all $\text{A1C}_2\text{H}_5$ is from the reaction $\text{A1CH}_2 + \text{CH}_3 = \text{A1C}_2\text{H}_5$ based on ROP analyses in 3C and 3C+MF. There are two major pathways for styrene ($\text{A1C}_2\text{H}_3$) formation both in 3C and 3C+MF experiments. A1CHCH_3 , from the H abstraction reaction of ethylbenzene, could undergo an H elimination reaction, forming $\text{A1C}_2\text{H}_3$ ($\text{A1CHCH}_3 = \text{H} + \text{A1C}_2\text{H}_3$). Another pathway is related to NPB radical ($\text{A1CHCH}_2\text{CH}_3$). $\text{A1CHCH}_2\text{CH}_3$ easily goes through β -C-C fission reaction to produce $\text{A1C}_2\text{H}_3$ ($\text{A1CHCH}_2\text{CH}_3 = \text{CH}_3 + \text{A1C}_2\text{H}_3$). The most important precursor in indene formation is the allylbenzene radical (A1CHCHCH_2) which can form indene via cyclization and dehydrogenation reaction ($\text{A1CHCHCH}_2 = \text{C}_9\text{H}_8 + \text{H}$). Moreover, 1-methylhydrindan radical ($\text{C}_9\text{H}_8\text{CH}_3$) is also an important precursor via reaction $\text{C}_9\text{H}_8 + \text{CH}_3 (+\text{M}) = \text{C}_9\text{H}_8\text{CH}_3 (+\text{M})$.

Figure 8 displays the detailed benzene formation pathways within the pyrolysis process of 3C and 3C+MF. ROP for benzene formation was conducted at the same temperature point as the sensitivity analysis. In 3C pyrolysis, about 20% benzene comes from the set of aromatics, while the above ratio in 3C+MF is about 14%, which could be caused by the higher concentration of aromatic species in 3C. There is no big gap for the total contribution of small molecules combination to benzene formation between in 3C+MF (57.4%) and 3C (55.6%). The combination of propargyl radical with allyl radical ($\text{aC}_3\text{H}_5 + \text{C}_3\text{H}_3 = \text{A1} + 2\text{H}$) has the highest contribution to the benzene formation in both inlet fuel mixtures. Except for the self-combination of C_3H_3 , the contribution of other small molecule reactions to benzene formation in 3C is higher than that in 3C+MF. It could be caused by the higher concentration of propyne, allene as well as allyl radical in 3C. It was found that C_3H_3 radical plays an important role in benzene formation. Almost all C_3H_3 radical is from the H abstraction reaction

of C_3H_4 (propyne and allene) during 3C pyrolysis. However, three important pathways can produce C_3H_3 radical in 3C+MF decomposition. More than half of the C_3H_3 radical is related to the MF pyrolysis procedure. Pathway 1 in Fig. 2 contributes about 35.9% of C_3H_3 radical; 17.4% of C_3H_3 radical comes from the unimolecular dissociation of 1-butyne which mainly is mainly generated by pathway 3, as shown in Fig. 2. The remaining C_3H_3 radical is from the H abstraction reaction of C_3H_4 . Non-benzenes cyclic hydrocarbons take part in more than 10% of benzene formation in total. Besides pathways shown in Fig. 8, fulvene, with H assistance, also has some contribution to benzene formation, occupying 4.4% and 6.4% respectively in 3C and 3C+MF.

As shown in Fig. 9, the naphthalene formation pathways perform no big difference between 3C and 3C+MF pyrolysis. A2 is mainly from the combination reaction of A1CH_2 with C_3H_3 ($\text{A1CH}_2 + \text{C}_3\text{H}_3 = \text{A2} + \text{H}$) rather than via HACA reaction, which is also the important naphthalene formation pathway in a toluene premixed laminar flame [58]. About 5.0% A2 is produced by styrene radical by reaction: $\text{A1CCH}_2 + \text{C}_2\text{H}_2 = \text{A2} + \text{H}$. 1,2-Dihydronaphthalenyl ($\text{C}_9\text{H}_7\text{CH}_3$) as well as its radical ($\text{C}_9\text{H}_7\text{CH}_2$) take part in more than 8% A2 generation in 3C and 3C+MF pyrolysis. The self-combination of cyclopentadiene radicals accounts for about 3.6% and 2.2% of A2 formation, respectively in 3C and 3C+MF. The reaction of benzene radical with C_4H_4 produces about 6.7% of A2 in 3C+MF, while it only contributes about 0.5% in 3C. It could be caused by the higher concentration of C_4H_4 in 3C+MF pyrolysis.

Figure 10 displays the comparison of carbon conversion rate between 3C and 3C+MF in view of the aromatic species. The carbon conversion rate generally increases with temperature rise. 3C+MF exhibits the lower tendencies to generate benzene at the same temperature point compared with 3C. There are two reasons for the higher concentration of aromatic species in 3C. One results from the higher concentration of NPB component which is the major source of aromatic species. Furthermore, given the carbon conversion rate in Fig. 6, it can be found that the carbon conversion during pyrolysis of 3C + MF has no significant difference to 3C from the point of light hydrocarbons. Hence, the carbon in oxygenated species, especially CO (occupying 4.7% carbon at 1223 K), could reduce the carbon ratio of aromatic species in 3C+MF mechanically. The lower tendencies to produce the set of aromatics indicate that MF, as a renewable fuel, may have better performance than traditional fuels in terms of soot reduction, which achieves good agreement with previous literature studies [7,8,10,15,28].

5. Conclusion

Pyrolysis experiment of 3C+MF was conducted in a flow reactor with a temperature range of 898–1223 K under atmospheric pres-

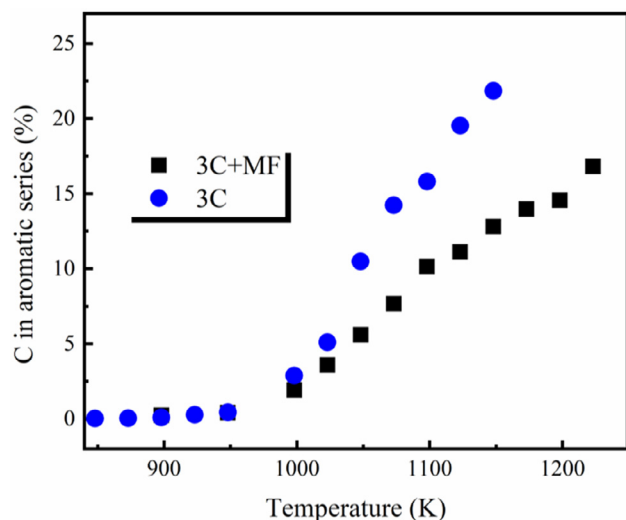


Fig. 10. Comparison of carbon conversion rate between 3C [31] and 3C+MF in view of aromatic species.

sure. A detailed kinetic model including 579 species and 3474 reactions was developed via validating against pyrolysis species profiles of 3C and 3C+MF with reasonable agreement. MF exhibits no obvious influence on the pyrolysis of 3C components decay behavior in contrast to 3C pyrolysis. When compared with pure MF decay, the pyrolysis of MF in blends leaning to low temperature, and the main decomposition pathway of MF in blends has been shifted from H migration reaction to H abstraction reaction at methyl side. The competition for radicals (especially CH_3 and H radicals), as well as longer residence time, play an important role in the consumption performance of fuel blends. The interaction of fuel blends was mainly through the common and important sensitive reactions which are also much to do with the CH_3 and H radicals. Except for acetylene and vinylacetylene, 3C pyrolysis has a higher concentration of light hydrocarbons than 3C+MF. Compared with 3C, 3C+MF could generate less aromatic species at the same temperature point, maximum difference could reach 10% at high temperature point, which could result from the higher concentration of NPB in the procedure of 3C pyrolysis as well as a portion of carbon coming into oxygenated species in 3C+MF decomposition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2021.111509.

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